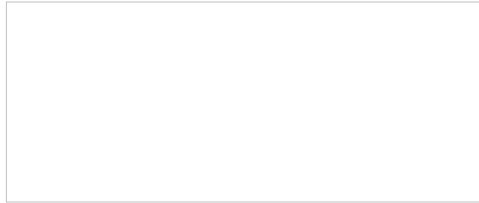


Pipeline Project Cost Estimation

June 8, 2025





Version: 1.0

Originated By:

Date:

Originator Electronic Signature:

Checked By:

Date:

Checked Electronic Signature:

Designer:

Date:

Designer Electronic Signature:

Project Name:

Document Number:

Revision Number:

Pipeline Cost Estimation

AACE Cost Estimate Classification

Level of Project Definition: Service, flow rate, Nominal Pipe Size, pipe wall thickness, coating or insulation requirements, length, route or terrain, numbers of various crossings and their lengths, and type/number of facilities shall be known to the user.

Accuracy range is dependent on the quality of information used to generate this estimate.

Methodology: Equipment cost is factored, parametric models are generated to calculate the cost of line pipe, coating, boring and construction.

End Usage: The typical purpose of this estimate should not be other than concept screening or feasibility studies.

Contact us at support@pipemath.com for further support.

Cost Estimation Calculations

Inputs

Nominal Pipe Size	2	Pipe OD	60.3 mm	>
External Coating	FBE	Internal Lining	Yes	
Transportation	Truck	Pipeline Length (Including Crossings)	1280 m	
Pipeline Wall Thickness	12.7 m	CP Test Station	\$ 3000 per each	
Ground Bed	\$ 25000 each			
Data Cable Cost	\$ 14 per m	Width of ROW	20 m	
HDD Crossing-Mud Motor	200 m	HDD Crossings-Mud Motor Unit Cost	150 per in-m	
Bored Crossing-Duckbill	50 m	Bored Crossing-Duckbill Unit Cost	35 per in-m	
Bored Crossing-TriHawk	0 m	Bored Crossing -TriHawk Unit Price	65 per in-m	
Open Cut Crossing (ROW, Roads)	30 m	Open Cut Crossings (ROW, Roads)	5000 per in-m	
Open Cut Crossing (Water body)	30 m	Open Cut Crossings (Water body)	10000 per in-m	
Equipment Selection	Pump			
Pump Rated Power	2500 kW			
Number of Station	1	Adjusted Pump (CAD)	11,666,667.00 \$	
Owner's Cost	10.00 %	Contractor's Cost To Support HDD	20.00 %	
		Steel Unit Cost	2900 \$ per ton	
		External Coating Unit Price	50 \$ per m ²	
		Internal Lining Cost	40 \$ per m ²	
		Transportaion Unit Cost	500 \$ per ton	
Shop Bends (Factor)	3 %	Pipeline Construction Unit Price	43.50 \$ per in-m	
Launcher/Receiver Station	2	Launcher/Receiver Station Unit Price	100,000.00 \$	
Block Valve Station	1	Block Valve Station Unit Price	100,000.00 \$	
Engineering Cost Pipeline	3 %			
Engineering Cost Facilities	10 %			
Mob/De-mob and indirect costs	5 %	ROW Logging	10000 \$ per hectare	
		ROW Preperation	2500 \$ per hectare	
		ROW Clearing	5000 \$ per hectare	
No. of Warning Sign	10	Warning Sign Unit Price	500 \$	
No. of Ditch Plugs	10	Ditch Plugs Unit Price	5000 \$	
No. of Diversion Berms	10	Diversion Berms Unit Price	5000 \$	
		Pipeline Cleaning, Hydrotesting and Batching	5 \$ per in-m	

Cost Estimation Calculations

Inputs

Construction Productivity Considerations:

Terrain

Level

Rock

0%

Productivity Factor

1

Length

0 m

Productivity Factor

1

Length

0 m

Cost Estimation Calculations

Summary

Description	Unit Price	Quantities	Cost
Steel Price	2,900.00 \$ per Ton	19.08 ton	55,339.87 \$
External Coating Cost	50.00 \$ per m ²	242.48 m ²	12,124.03 \$
Internal Lining Cost	40.00 \$ per m ²	140.34 m ²	5,613.65 \$
Transportation Cost (Truck)	500.00 \$ per Ton	19.08 ton	9,541.36 \$
Shop Bends	3.00 %	Factored	2,478.57 \$
Cathodic Protection Cost			28,000.00 \$
Data Cable Cost	14.00 \$ per m	1,280.00 m	17,920.00 \$
Total Pipeline Material Cost			131,017.48 \$
Pipeline Construction Cost	43.50 \$ per in-m	1,280.00 m	111,360.00 \$
Total Productivity Factor Cost		0.00 m	0.00 \$
Total Trenched Construction Cost			111,360.00 \$
HDD Crossings-Mud Motor	150.00 \$ per in-m	200.00 m	60,000.00 \$
Bored Crossing-Duckbill	35.00 \$ per in-m	50.00 m	3,500.00 \$
Bored Crossing-TriHawk	65.00 \$ per in-m	0.00 m	0.00 \$
Open Cut Crossings (ROW, Roads)	5,000.00 \$ per in-m	30.00 m	300,000.00 \$
Open Cut Crossings (Waterbody)	10,000.00 \$ per in-m	30.00 m	600,000.00 \$
Pipeline Contractor Support for HDD	20.00 %		12,000.00 \$
Total Crossing Cost			975,500.00 \$
Pump Station Factored Cost	1.00 units	2500 kW	11,666,667.00 \$
Launcher/Receiver Station	2.00 units	100,000.00 \$	200,000.00
Block Valve Station	1.00 units	100,000.00 \$	100,000.00
Total Cost			1,517,877.48 \$
Engineering Cost Pipeline	3.00 %	Factored	36,536.32 \$
Engineering Cost Facilities	10.00 %	Factored	1,196,666.75 \$
Other Costs			
Mob/De-mob and indirect costs	5.00 %	Factored	54,343.00 \$
ROW Clearing	5,000.00 \$ per hectare		12,800.00 \$
ROW Preparation	2,500.00 \$ per hectare		6,400.00 \$
ROW Logging	10,000.00 \$ per hectare		25,600.00 \$
Warning Signs	10.00 units		5,000.00 \$
Ditch Plugs	10.00 units		50,000.00 \$
Diversion Berms	10.00 units		50,000.00 \$
Pipeline Cleaning, Hydrotesting and Batching	5.00 \$ per in-m		12,800.00 \$
Total Other Pipeline Cost			216,943.00 \$

Cost Estimation Calculations

Results

Results:

Description	Unit Price	Quantities	Cost
Pipeline Construction Cost Level Terrain	43.50 \$ per in-m	0.00 m	0.00 \$
Pipeline Construction Cost 0% Rock	43.50 \$ per in-m	0.00 m	0.00 \$

Cost Estimation Calculations

Estimation Summary

Description	Unit Price	Quantities	Cost
Total Pipeline Material Cost			131,017.48 \$
Total Trenched Construction Cost			111,360.00 \$
Total Crossing Cost			975,500.00 \$
Total Other Pipeline Cost			216,943.00 \$
Owner's Cost			147,135.68 \$
Total Pipeline Cost			1,618,492.48 \$
Engineering Cost Pipeline	3.00 %	Factored	36,536.32 \$
Pipeline Cost (in-m)			632.22 \$
Pump Station Factored Cost	1.00 units	2500 kW	11,666,667.00 \$
Launcher/Receiver Station	2.00 units	100,000.00 \$	200,000.00
Block Valve Station	1.00 units	100,000.00 \$	100,000.00
Engineering Cost Facilities	10.00 %	Factored	1,196,666.75 \$
Owner's Cost			1,316,333.38 \$
Total Facilities Cost			14,479,667.00 \$
Overall Project Cost			16,098,159.48 \$